

INSTITUTE OF ENGINEERING AND MANAGEMENT, KOLKATA

Department of Computer Science & Engineering

LABORATORY ASSIGNMENT REPORT ASSIGNMENT 1: BASIC SYNTAX AND KEYWORDS

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Department	:	CSE (AI & ML)
Course	:	Python for Automation
Academic Session	:	2023-2027
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1. MODULE NAME & EXPERIMENT TITLE

Module Name: Test Automation Frameworks (Robot Framework)

Experiment Title: Basic Syntax and Keywords in Robot Framework

2. PROBLEM STATEMENT

Design and implement test scripts using Robot Framework to automate web browser interactions. Specifically, the assignment requires solving three tasks:

- Create a Robot Framework test case to open a web browser and navigate to a specific URL.
- Use the "Input Text" keyword to fill in a form field on a webpage.
- Write a test case to verify the presence of a specific element on a webpage using the "Page Should Contain Element" keyword.

3. OBJECTIVE

- To understand the basic syntax structure of standard Robot Framework files (Settings, Variables, and Test Cases sections).
- To learn how to integrate SeleniumLibrary with Robot Framework for GUI browser automation.
- To practice browser initialization, form input handling, and element assertions in web pages.

4. TOOLS, SOFTWARE, AND CONCEPTS USED

- **Programming Language / Environment:** Python 3.11.x
- **Framework:** Robot Framework 6.1
- **Library:** robotframework-seleniumlibrary
- **Browser & Driver:** Google Chrome & ChromeDriver
- **IDE / Code Editor:** Visual Studio Code (VS Code) with Robot Framework Language Server extension
- **Key Concepts:** Keyword-driven testing, tabular syntax formatting (minimum 2 spaces between arguments), web element locators (ID, Name, XPath), DOM assertions.

5. SOURCE CODE / IMPLEMENTATION STEPS

The code is saved in a file named `assignment1_tests.robot`. VS Code was used to write the script adhering strictly to Robot Framework's tabular spacing rule (4 spaces used as delimiter between keywords and arguments).

```
assignment1_tests.robot - Python for Automation Robot Framework

1  *** Settings ***

2  Documentation Assignment 1 - Basic Syntax and Keywords
```

6. OUTPUT WITH LABELLED TERMINAL & EXECUTION SCREENSHOTS

The test suite was executed from the Integrated Terminal in VS Code using the command `robot assignment1_tests.robot`. Below are the execution logs and terminal output.

Screenshot 1: VS Code Terminal Output Execution Window

```
PS C:\Users\Ashmita\Automation_Lab> robot assignment1_tests.robot
===== Assignment1
Tests ===== TC_01:
Open Browser And Navigate To Specific URL | PASS | -----
----- TC_02: Fill In Form Field Using Input Text Keyword |
PASS | ----- TC_03:
Verify Element Presence Using Page Should Contain Element | PASS |
===== Assignment1
Tests | PASS | 3 tests, 3 passed, 0 failed
===== Output:
C:\Users\Ashmita\Automation_Lab\output.xml Log: C:\Users\Ashmita\Automation_Lab\log.html
Report: C:\Users\Ashmita\Automation_Lab\report.html PS C:\Users\Ashmita\Automation_Lab>
```

Screenshot 2: Generated HTML Log Verification (Report Summary)

SUCCESS: All 3 Test Cases Executed and Passed

Test Status:	3 Passed, 0 Failed
Elapsed Time:	00:00:04.128
Executed Keywords:	Open Browser, Maximize Browser Window, Input Text, Page Should Contain Element, Close Browser

7. BRIEF RESULT & OBSERVATIONS

- **Result:** All 3 test cases in the test suite passed successfully without any locator timeouts or syntax errors.
- **Observations:**
 - Robot Framework strictly requires proper separation between keywords and arguments (at least 2 consecutive spaces). Using a single space results in argument parsing failure.
 - The Input Text keyword correctly located the target input fields using standard locator syntax (`id=username`) and entered the string value without requiring additional clear keys.
 - The Page Should Contain Element keyword verified DOM element presence accurately; if an invalid locator is supplied, Robot Framework raises an `'ElementNotFound'` exception and fails the test step.

8. CONCLUSION

In this laboratory experiment, I successfully learned the fundamental structure of Robot Framework test scripts. I gained practical experience in initializing browser sessions using `SeleniumLibrary`, interacting with web forms using `Input Text`, and asserting DOM elements with `Page Should Contain Element`. The experiment demonstrated the efficacy and simplicity of keyword-driven testing for browser automation.